

Design Verification Plan and Report (virtual test) —

VBF-T10A-DVPR-01

Virtual-test version. Values from simulation output; uncovered conditions marked N/A.

Item	Condition	Result value	Determination	Source
1C discharge capacity	1C, 298.15 K	5.057 Ah	✓ (nominal 5.0 Ah)	`cell/T10_1c_dfn.json:capacity_ah`
Energy density	contract caliber	426.89 Wh/kg	✓ (≥ 392.61)	`cell/T10_energy_dfn.json:energy_density_wh_kg`
4C fast-charge temperature rise	4C, 318.15 K, lumped thermal	T_max 324.55 K	✓ ($\leq 333.15\text{ K} = 60\text{ }^{\circ}\text{C}$)	`cell/T10_4c_dfn.json:T_max_K`
4C fast-charge plating	4C + plating module	anode min +0.0341 V	✓ (plated = false)	`cell/T10_4c_dfn.json:anode_potential_v`
Overcharge → thermal runaway	0.5C charge to 4.70 V + run-tr ODE	triggered = false	✓	`cell/T10_tr.json:triggered`
Voltage window	parameter set	2.5 – 4.2 V	✓	parameter set
Nail penetration	—	—	N/A (beyond simulation boundary, requires physical experiment)	—
Crush / drop	—	—	N/A	—
Cycle life	aging	—	N/A (aging model not run for this chemistry)	—
Rate-pulse internal resistance	—	—	N/A (DCR 0.0185 Ω from 1C discharge available as proxy)	`cell/T10_energy_dfn.json:dcr_ohm`

Conclusion

****Pass**** on all task criteria: energy density ≥ 392.61 Wh/kg (426.89), 4C no plating (anode +0.0341 V), T_max $\leq 60\text{ }^{\circ}\text{C}$ (51.4 $^{\circ}\text{C}$), overcharge 4.7 V no thermal runaway (triggered=false).

Uncovered (paper limitations): nail penetration, crush, drop, cycle life, full temperature range — all beyond pure-simulation boundary.